

CATEGORY	EXCEEDS EXPECTATIONS (76-100%)	MEETS EXPECTATIONS (51-75%)	PARTIALLY MEETS EXPECTATIONS (26-50%)	DID NOT MEET EXPECTATIONS (0-25%)	TOTAL
TEAM PROJECT PROPOSAL PRESENTATION	Proposal clearly and succinctly articulates the project's intent, goals, and the AWS technologies to be used.	Proposal states the project's intent and goals but includes minor issues with clarity or AWS technology integration.	Proposal outlines the project but is unclear about the goals or lacks details about the AWS technologies.	Proposal is vague, missing key details about the project's goals and AWS technologies.	5
TEAM CHECK-INS	See Project Check-In document in myCourses, which details deliverables for each of the 3 check-ins				20
USER'S GUIDE	User's guide is clear, complete, and includes all necessary instructions for project setup and use.	User's guide is clear but may have minor ambiguities around setup or usage.	User's guide is incomplete or unclear, causing confusion about setup or usage.	User's guide is absent or fails to provide actionable instructions.	10
FINAL PRESENTATION	Presentation clearly defines the problem and solution, includes cost analysis, and showcases a polished, fully functional demo.	Presentation defines the problem and solution well but includes minor clarity issues or a demo with small technical hiccups.	Presentation lacks clarity, misses key details about the problem or solution, or the demo encounters significant technical problems.	Presentation is unclear or incomplete, and the demo is either missing or fails to function effectively.	20
PROJECT SETUP & TEARDOWN	Project setup and teardown are automated using GitHub Actions, executing seamlessly with no issues. Documentation is clear and complete, enabling automated workflows without assistance.	Project setup and teardown using GitHub Actions run successfully with minor, easily resolved issues. Documentation is mostly clear with a few needed clarifications.	Significant issues occurred during project setup or teardown using GitHub Actions, requiring team assistance. Documentation was incomplete or unclear, leading to confusion.	Project setup or teardown failed, or GitHub Actions was not utilized as required. Documentation was missing or lacked clarity, requiring substantial team intervention.	20
OVERALL EFFORT, DESIGN & EXECUTION	The team used all 5 required AWS technologies and integrated at least one additional AWS service in a meaningful way. The project demonstrated strong technical depth, thoughtful design, and went beyond the assignment's core requirements. Execution was complete, polished, and showed clear evidence of extra effort and innovation.	The team used all 5 required AWS technologies and may have included an additional service. The solution met assignment requirements with acceptable complexity and effort, though it remained within standard expectations. Execution was mostly complete, with only minor gaps or missed opportunities for enhancement.	The team used all 5 required AWS technologies, but the solution lacked expected complexity, depth, or integration. Implementation showed limited effort, minimal technical design, or noticeable gaps in functionality. Project goals were only partially met.	The team used 5 or fewer required AWS technologies, but the solution was either incomplete, poorly integrated, or lacking in complexity. Significant execution issues were present, and the project did not meet core objectives or reflect sufficient effort.	25
TOTAL POINTS					100

Due 12/1
Presentation
12/1 & 12/3

Remaining
project
deliverables
due on 12/8 @
midnight

- ▶ Which of the following best describes batch processing?
 - A. Processes data continuously with very low latency
 - B. Handles individual records one at a time in real-time
 - C. Processes large sets of collected data with latency in minutes or hours ☒
 - D. Only works with structured relational data

- ▶ In streaming systems, the value of data decreases if it is not processed quickly
 - ▶ True ☒
 - ▶ False

- ▶ Spark Streaming processes data purely as continuous streams without batching
 - ▶ True
 - ▶ False ☒

- ▶ Up to this point, you've had a crash course on the Hadoop ecosystem and some key tools that you can use to process Big Data problems



- ▶ Using AWS EMR, you can create Hadoop clusters that process data of literally any size
- ▶ Some of the key benefits to note:
 - ▶ Ease of use – Launch a fully running cluster in minutes
 - ▶ Low cost – A 10-node cluster can run for as little as .15/hour
 - ▶ Reliable – EMR is tuned for the cloud, requiring less time tuning and monitoring
 - ▶ Security – Several security features to keep data safe
 - ▶ Flexible – Full control over the cluster
- ▶ This has led more organizations to leverage the cloud for Big Data Analytics by creating “Data Lakes”



Big Data on the Cloud: Data Lakes & Analytics

Engineering Cloud Software Systems

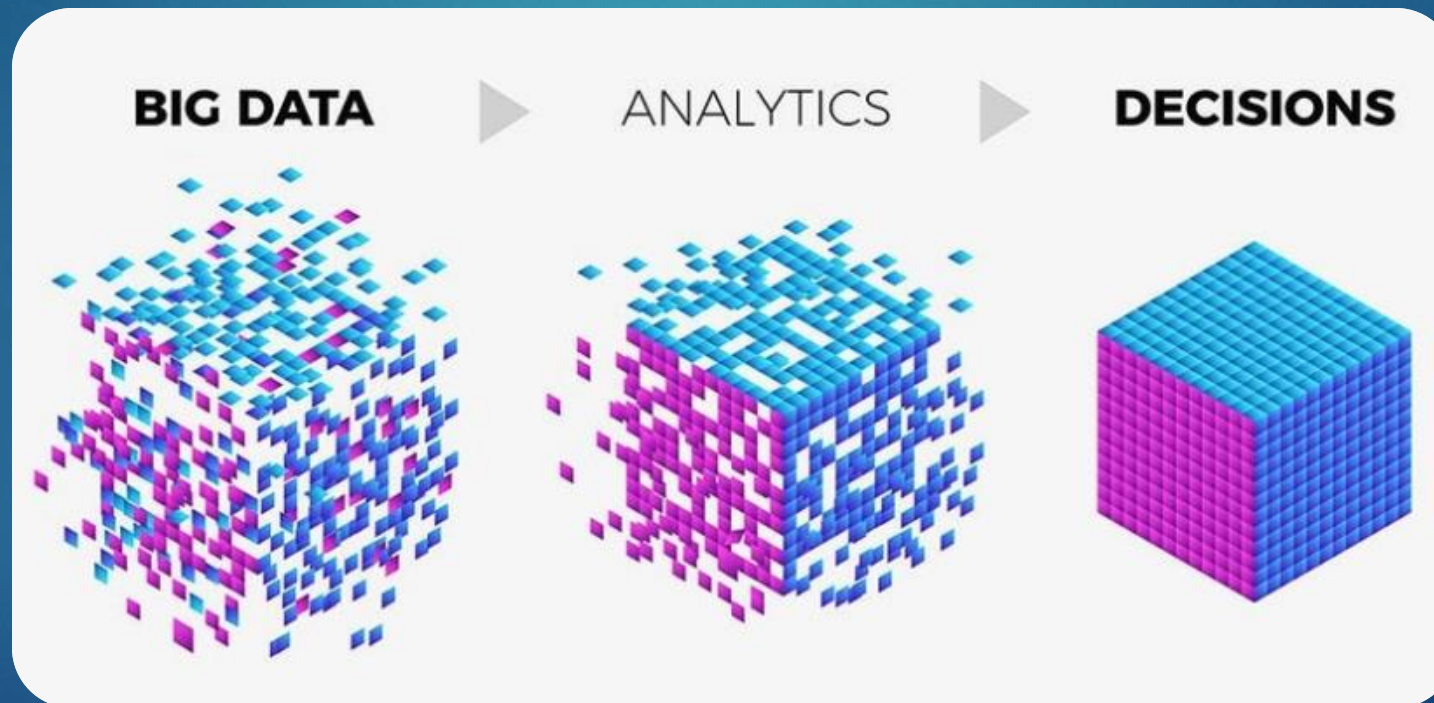
Department of Software Engineering
Rochester Institute of Technology



Big Data Analytics

10

- Big Data Analytics is like searching for meaningful patterns in an ocean of information using powerful tools to spot the hidden currents, trends, and signals that guide smarter decisions



Big Data Analytics – Why Important?

11

► Cost reduction

- Big data technologies bring significant cost advantages when it comes to storing large amounts of data – plus they can identify more efficient ways of doing business

► Faster, smarter decisions

- Tools like Hadoop and in-memory analytics enable real-time analysis of diverse data sources, allowing organizations to act quickly and confidently

► Innovation and Personalization

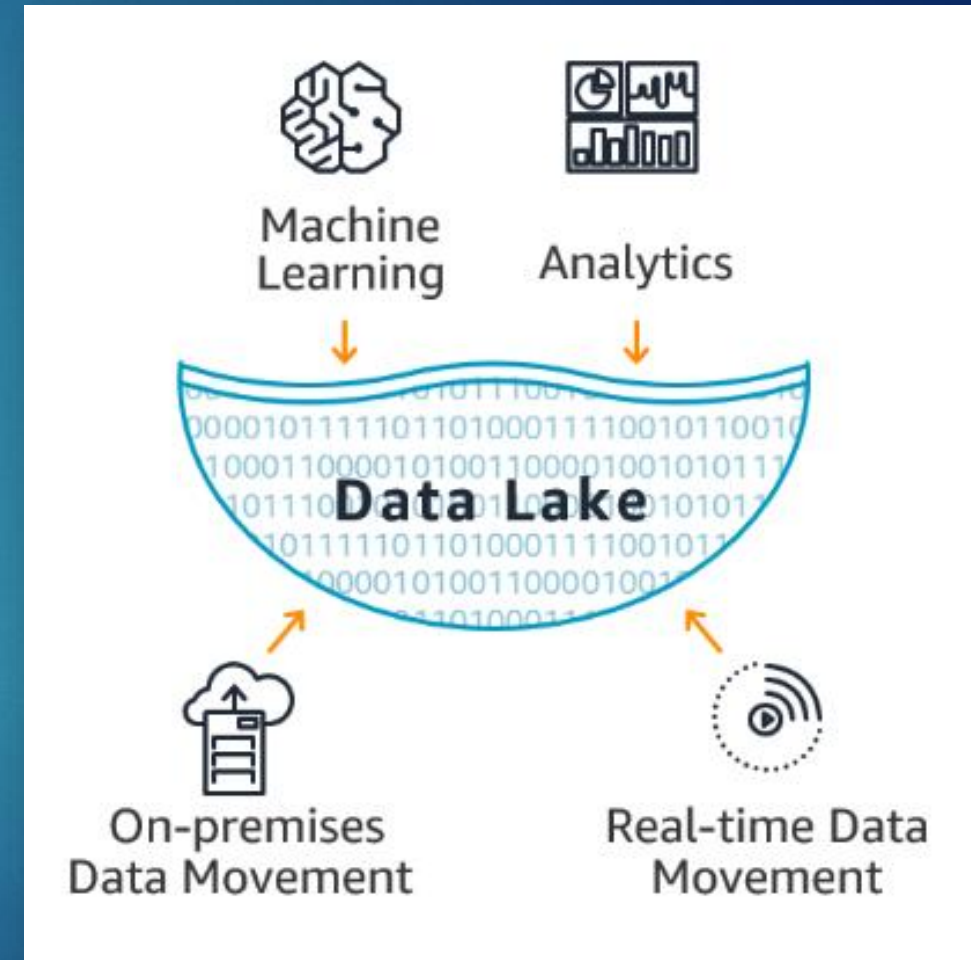
- Analytics help businesses understand customer needs and satisfaction, empowering them to create new products, services, and tailored experiences



Data Lake - Defined

12

- ▶ A Data Lake is a centralized repository that can store structured, semi-structured, and unstructured data at any scale
- ▶ It typically serves as a single source for all enterprise data, holding both raw source data and transformed data for reporting, visualization, analytics, and machine learning
- ▶ Unlike traditional databases, a Data Lake allows you to store data as-is without predefined structure so it can be explored and used for future needs



Data Lake – How do they work?

13

The concept can be compared to a water body, a lake, where water flows in, filling up a reservoir and flows out.

STRUCTURED DATA

1. Information in rows and columns
2. Easily ordered and processed with data mining tools

1

The incoming flow represents multiple raw data archives ranging from emails, spreadsheets, social media content, etc.

2

The reservoir of water is a dataset, where you run analytics on all the data.

3

The outflow of water is the analyzed data.

4

Through this process, you are able to “sift” through all the data quickly to gain key business insights.

UNSTRUCTURED DATA

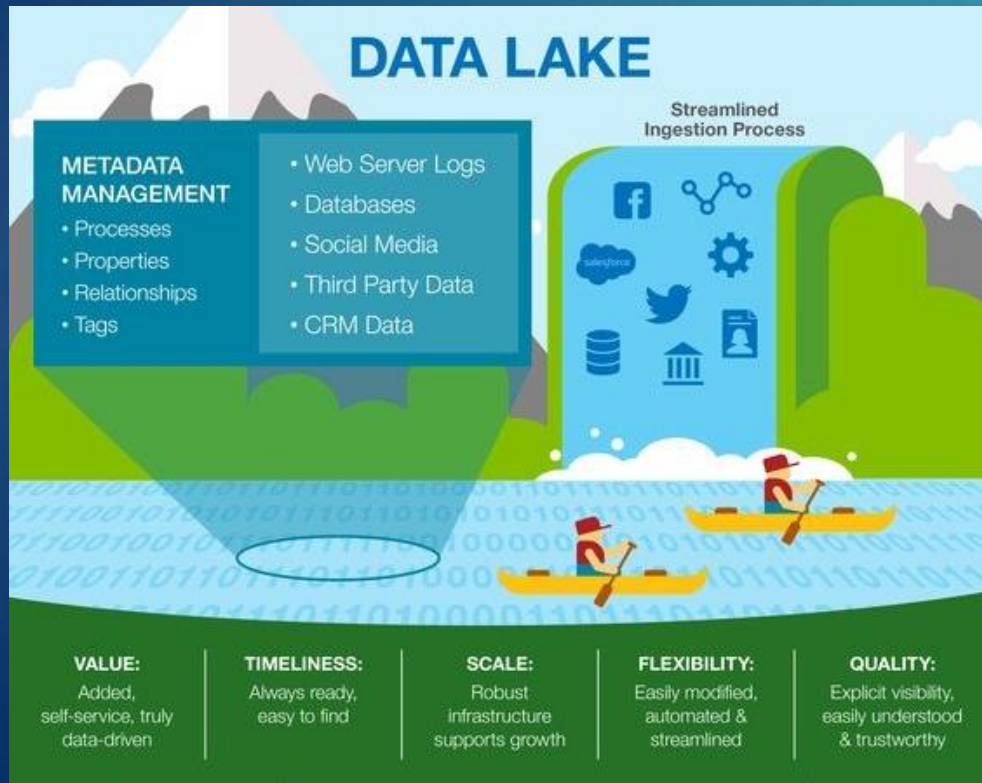
1. Raw, unorganized data
2. Emails
3. PDF files
4. Images, video and audio
5. Social media tools



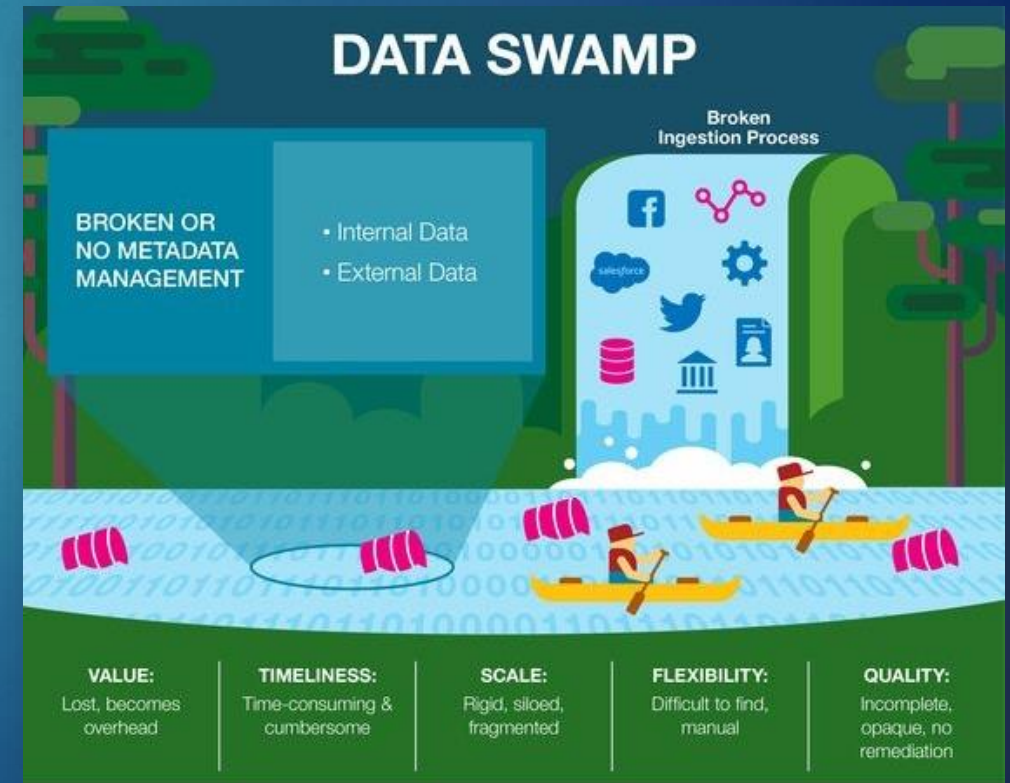
Data Lake - Challenges

14

- ▶ A main challenge with a Data Lake architecture is that raw data is stored with no oversight of the contents (i.e. it's a dumping ground of anything and everything)
- ▶ For a Data Lake to make data usable, it needs to have clearly defined mechanisms to cataloging, curation, and securing data
- ▶ Without these elements, data cannot be found, or trusted resulting in a “Data Swamp”



VS.

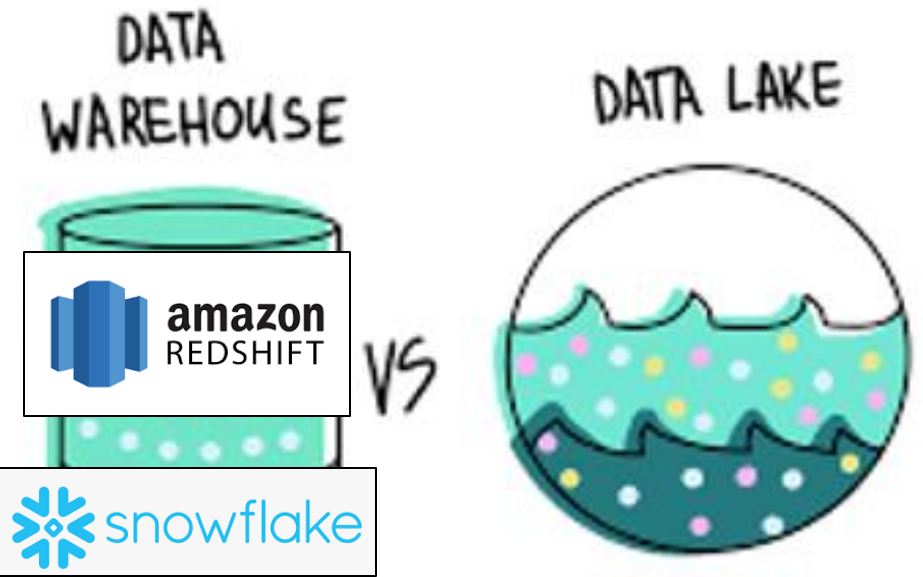


Data Lake vs. Data Warehouse

15

- ▶ Data Lakes and Data Warehouses are both widely used for storing big data, but they are not interchangeable terms
- ▶ A Data Warehouse is a repository for structured, filtered data that has already been processed for a specific purpose
- ▶ A Data Lake is a vast pool of raw data, the purpose for which is not yet defined

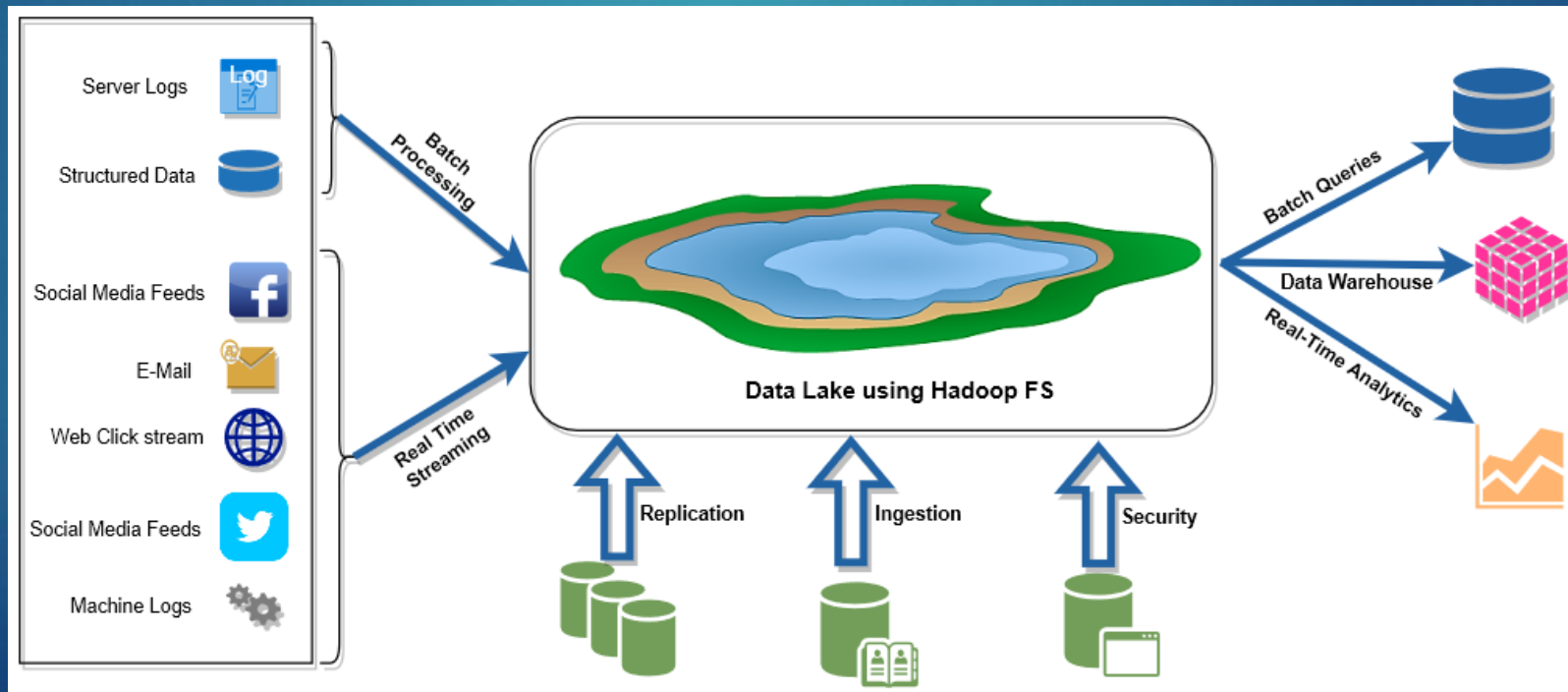
DATA WAREHOUSE	vs.	DATA LAKE
structured, processed	DATA	structured / semi-structured / unstructured, raw
schema-on-write	PROCESSING	schema-on-read
expensive for large data volumes	STORAGE	designed for low-cost storage
less agile, fixed configuration	AGILITY	highly agile, configure and reconfigure as needed
mature	SECURITY	maturing
business professionals	USERS	data scientists et. al.



Data Lake → Hadoop

16

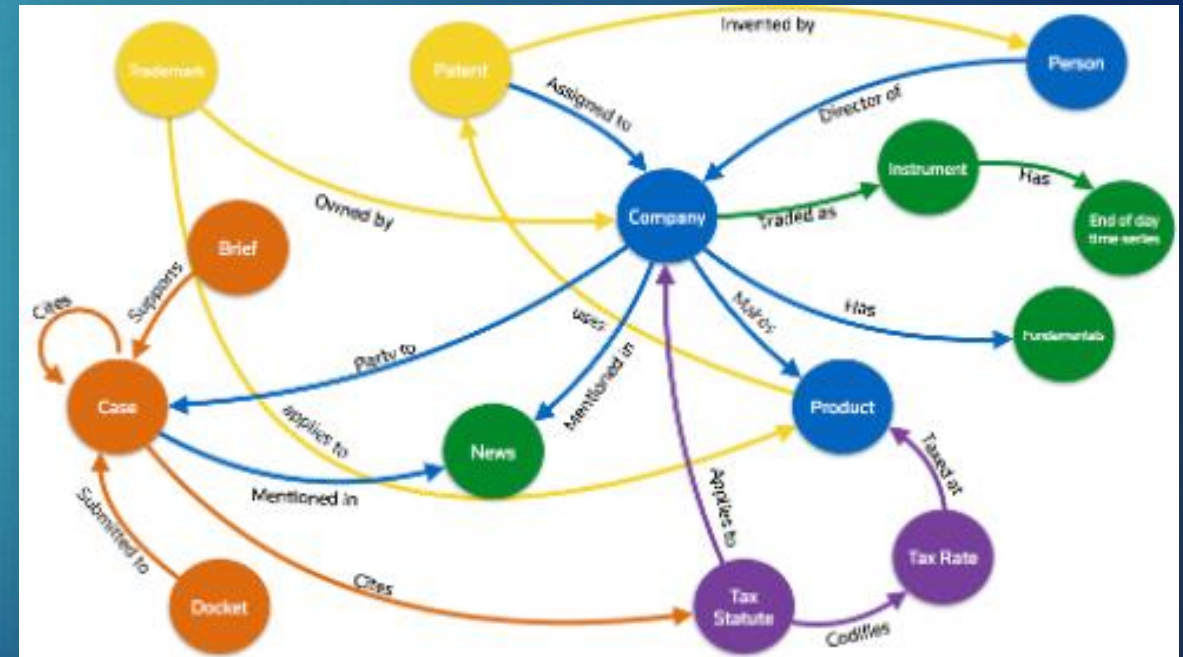
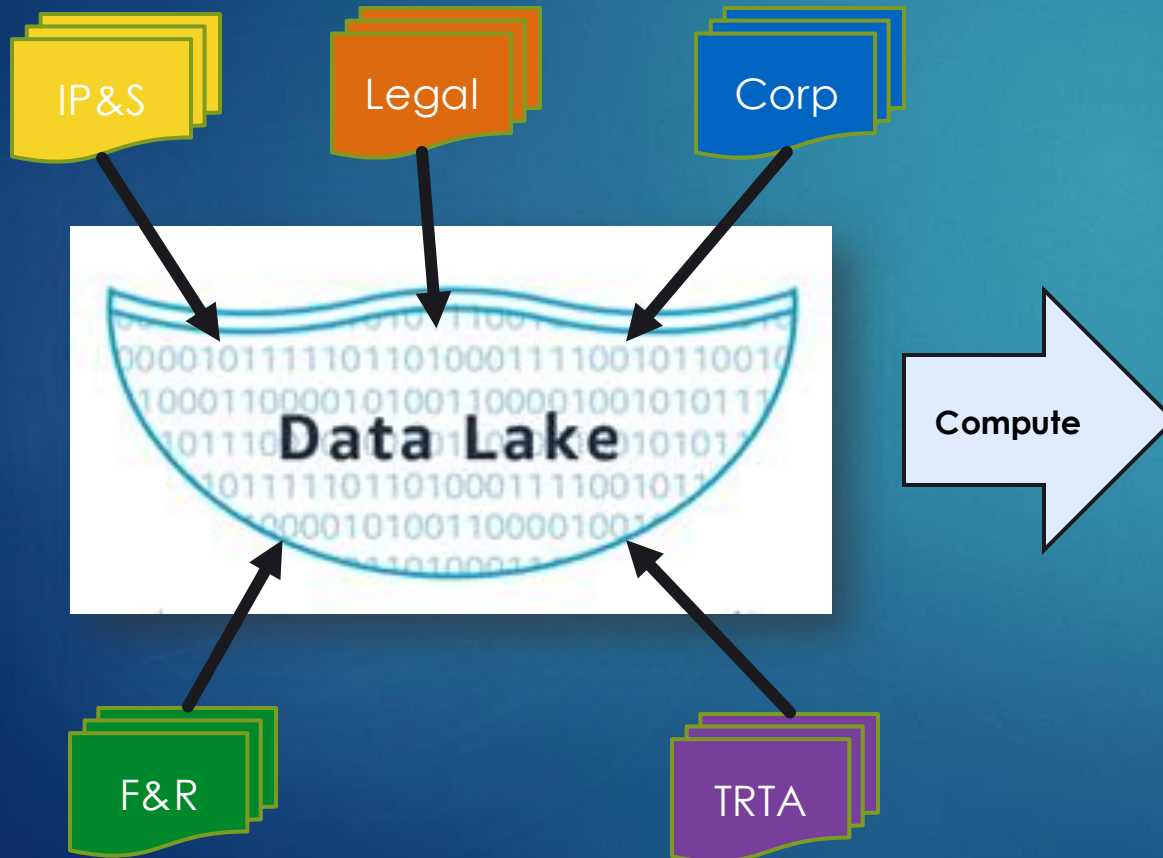
- ▶ Hadoop provided the foundation for Data Lake architectures as it provides a cheap way for organizations to store all their data on commodity hardware
- ▶ The term Data Lake is often associated with Hadoop-oriented storage HDFS
 - ▶ In such a scenario, an organization's data is first loaded into the Hadoop platform, and then business analytics and data mining tools are applied to the data where it resides on Hadoop's cluster nodes of commodity computers



Data Lake Example...

17

- ▶ Business Units (Legal, IP&S, F&R, etc.) all had their own content they managed
- ▶ The “Knowledge Graph” would connect all this content in ways that was previously not possible

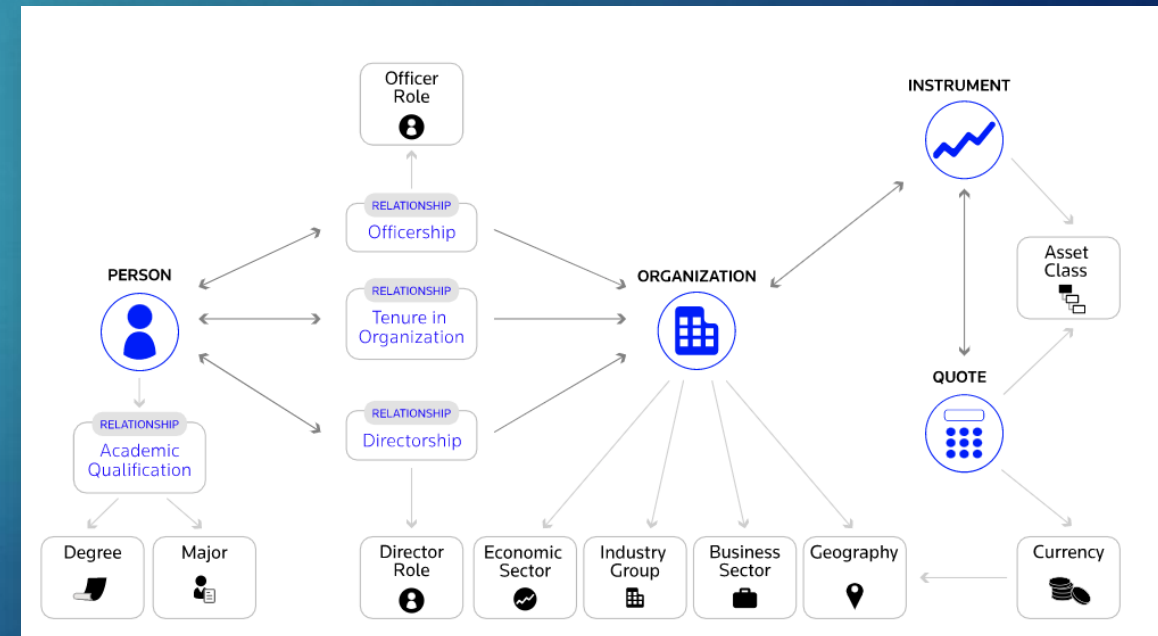
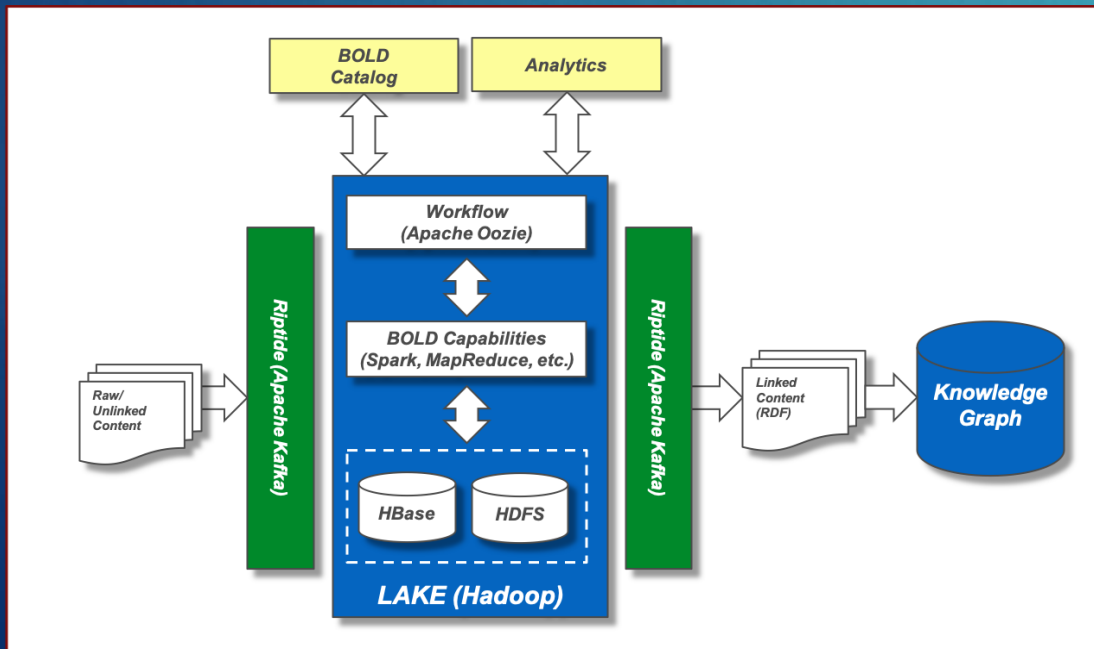


Knowledge Graph

Data Lake Example...

18

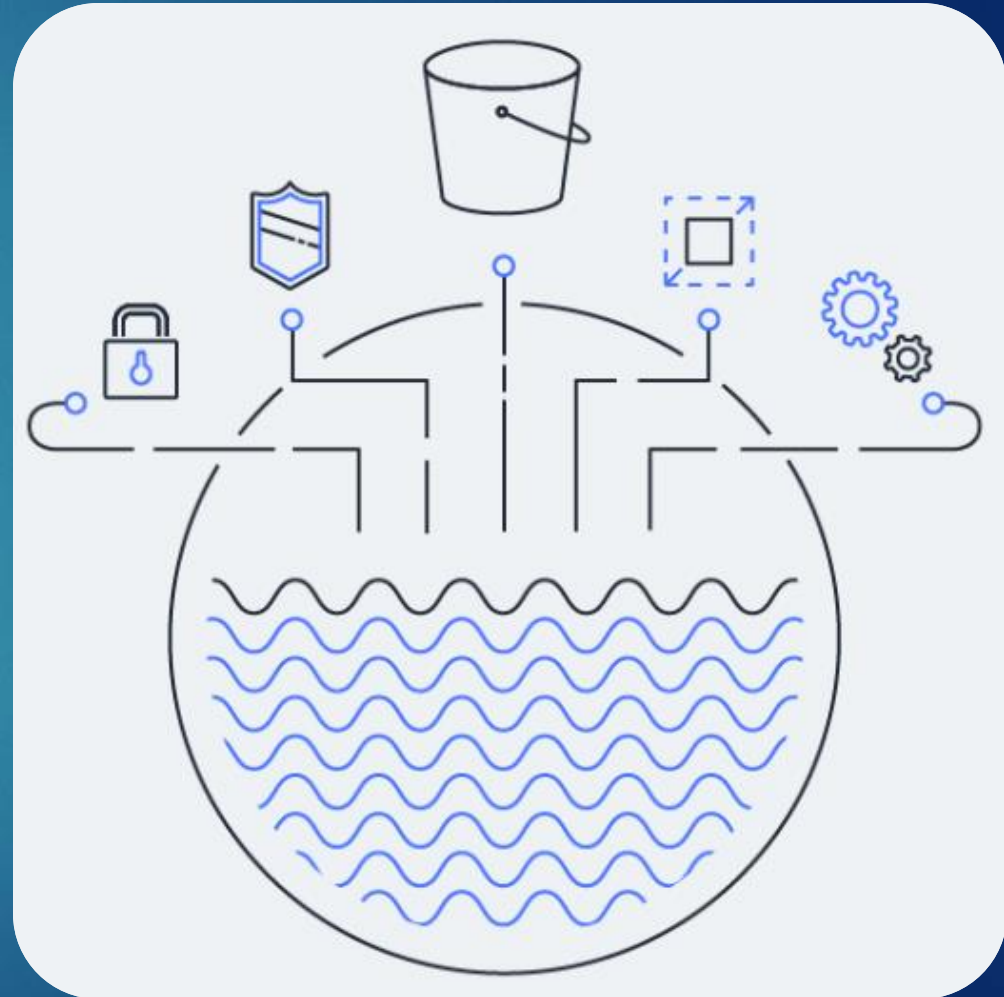
- ▶ PermlD.org
- ▶ Built on premise as Project BOLD (Big Open Linked Data)
- ▶ Technologies
 - ▶ Spark, MapReduce, HBase, Kafka, Oozie and Cassandra (Knowledge Graph)



Data Lakes in the cloud on AWS

19

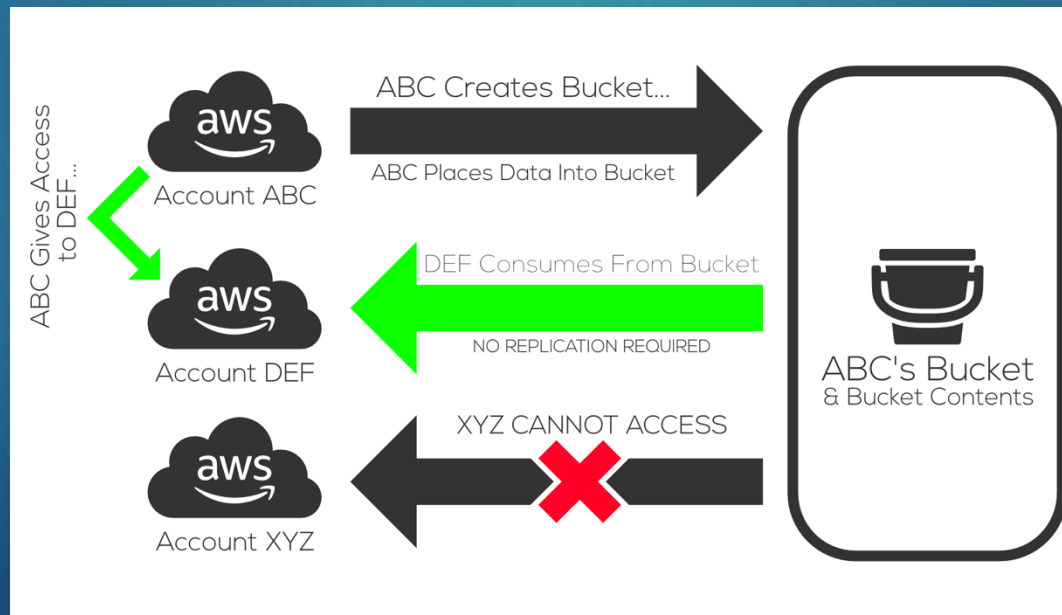
- ▶ With a Data Lake built on Amazon S3, you get extreme durability (99.999999999%)
- ▶ There are several native AWS services to run Big Data analytics, artificial intelligence, machine learning to gain insights from your unstructured data sets
- ▶ You can automatically scale up storage and processing capacity, without lengthy resource procurement cycles
- ▶ Can you identify another advantage with S3?



Data Lakes in S3 = Content Sharing

20

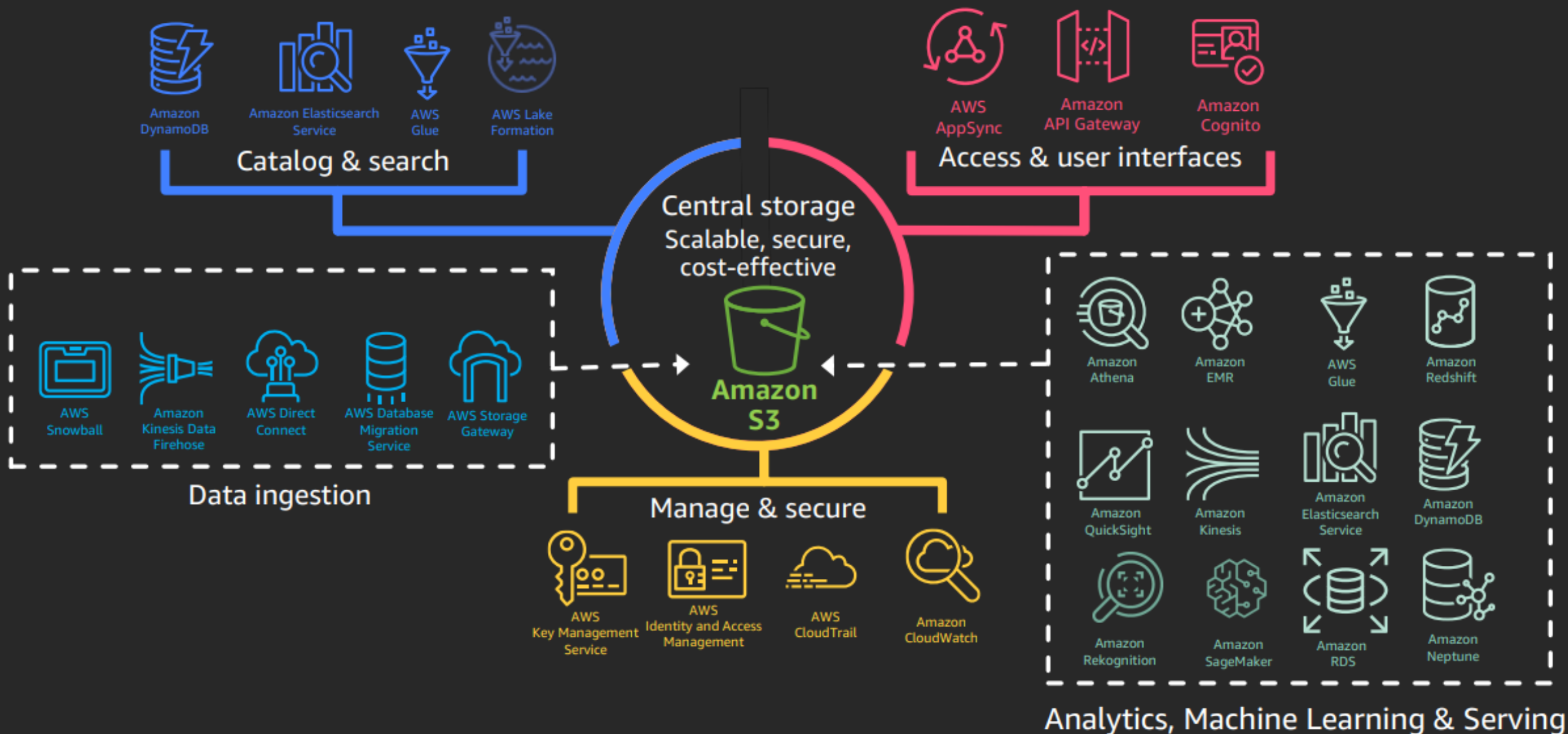
- ▶ With the the right credentials, you can see the contents of another company's S3 bucket(s) right from your own personal AWS account
- ▶ No physical copying required, which is a major shift in mindset from how companies do things in the on-premise world
- ▶ Where we focus our time on isolating data with physical infrastructure, cloud computing shifts are attention to focus on isolating data using security policies



More on this later...

Data Lake on AWS

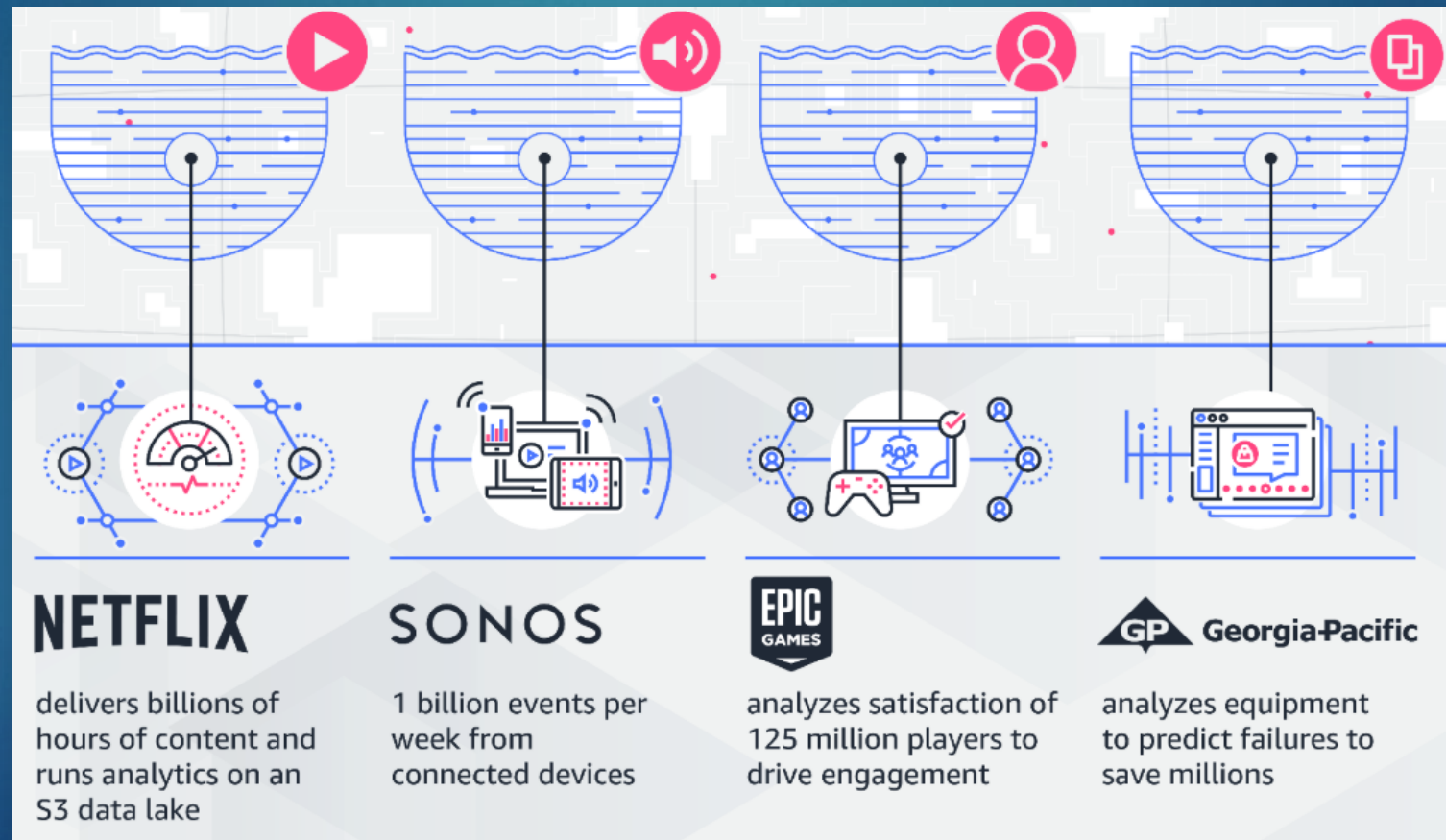
21



Who uses Data Lakes on AWS?

22

- ▶ There are 700,000 data lakes run on AWS
- ▶ A few examples...



Data Visualization

23

- ▶ Data Visualization is the practice of presenting data in pictorial or graphical formats such as charts, graphs, and dashboards
- ▶ It helps decision makers quickly understand complex information and uncover patterns, trends, and insights that may be hard to see in raw data.
- ▶ Interactive visualization allows users to drill down, filter, and explore data dynamically to reveal deeper insights in real time



Data Exploration & Visualization Tools on AWS

24

▶ Amazon QuickSight

- ▶ As a fully managed service, you can create and publish interactive dashboards that include ML Insights

▶ Amazon Sagemaker and EMR Notebooks

- ▶ A managed environment based on Jupyter Notebooks that allows data scientists, analysts, and developers to prepare and visualize data, collaborate with peers, build applications, and perform interactive analysis

▶ Amazon Athena

- ▶ An interactive query service that's capable of seamlessly using standard Structured Query Language (SQL) or Spark SQL
- ▶ First, we need to understand a little bit about Spark SQL...

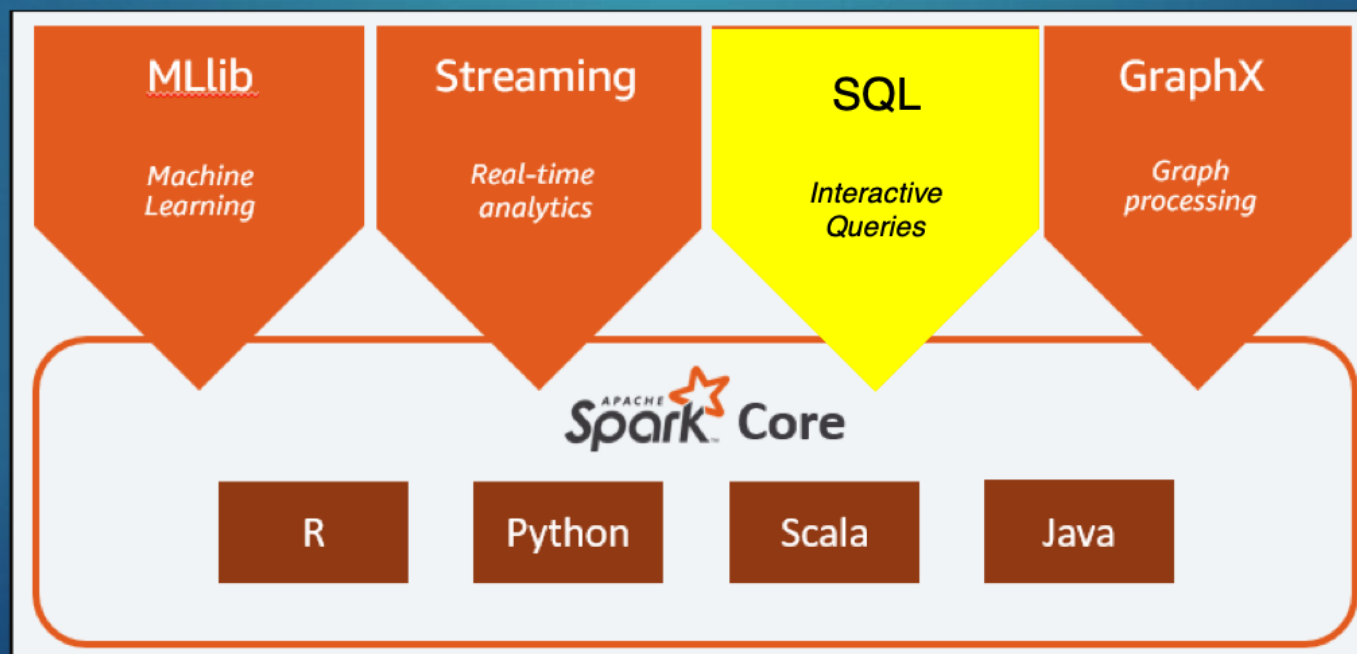


Amazon Athena

Spark SQL

25

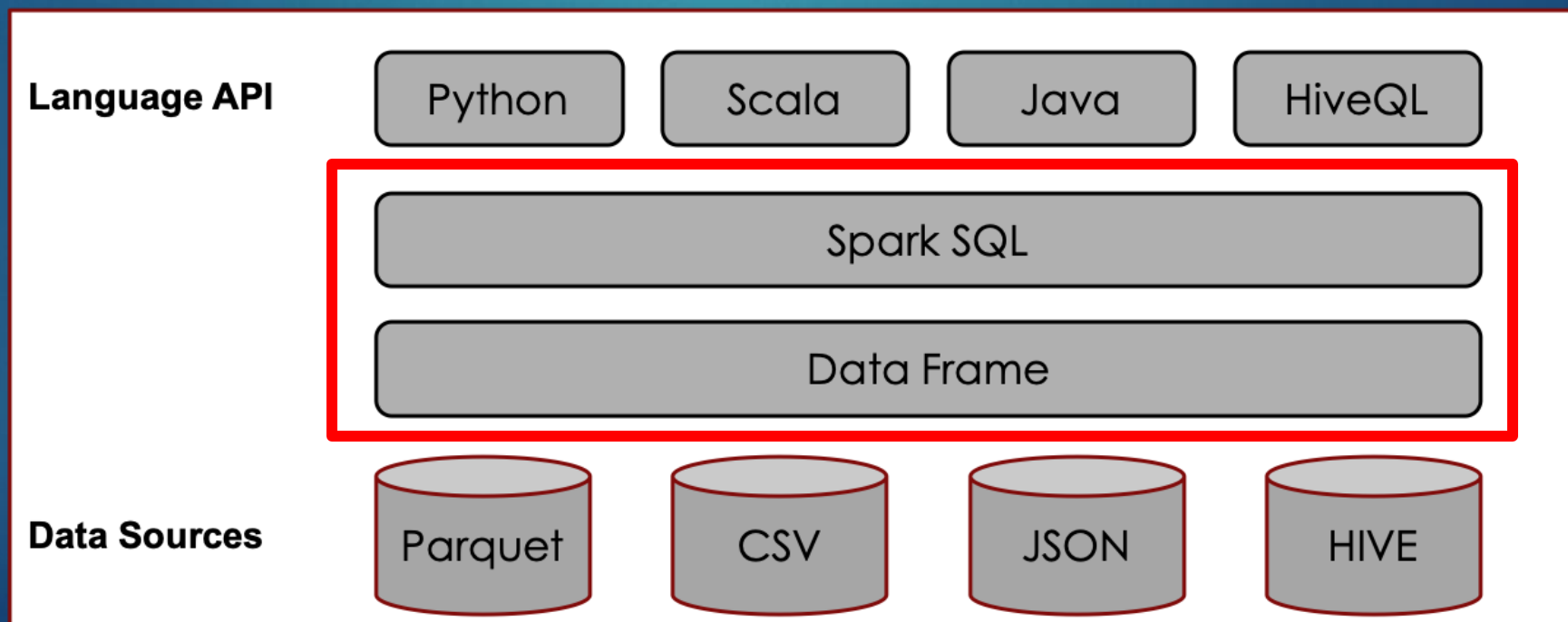
- ▶ Spark SQL is Spark's interface for working with structured data
- ▶ Those familiar with RDBMS can easily relate to the syntax of Spark SQL
- ▶ It was developed to address some of the shortcomings of Hive (e.g. performance of MapReduce)



Spark SQL - Features

26

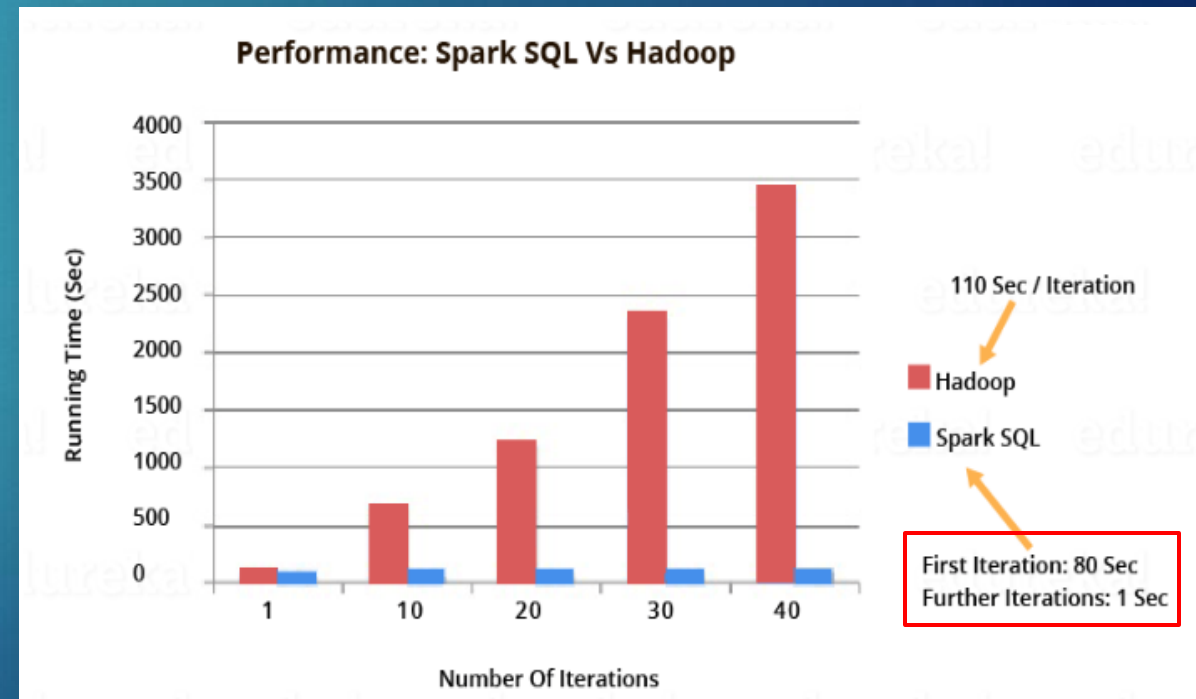
- ▶ Spark SQL queries are integrated with Spark programs. It allows us to query structured data inside Spark programs, using SQL or a DataFrame API which can be used in Java, Scala, Python and R
- ▶ Both DataFrames and SQL support a common way to access a variety of data sources, like Hive, Parquet, JSON, and JDBC



Spark SQL - Features

27

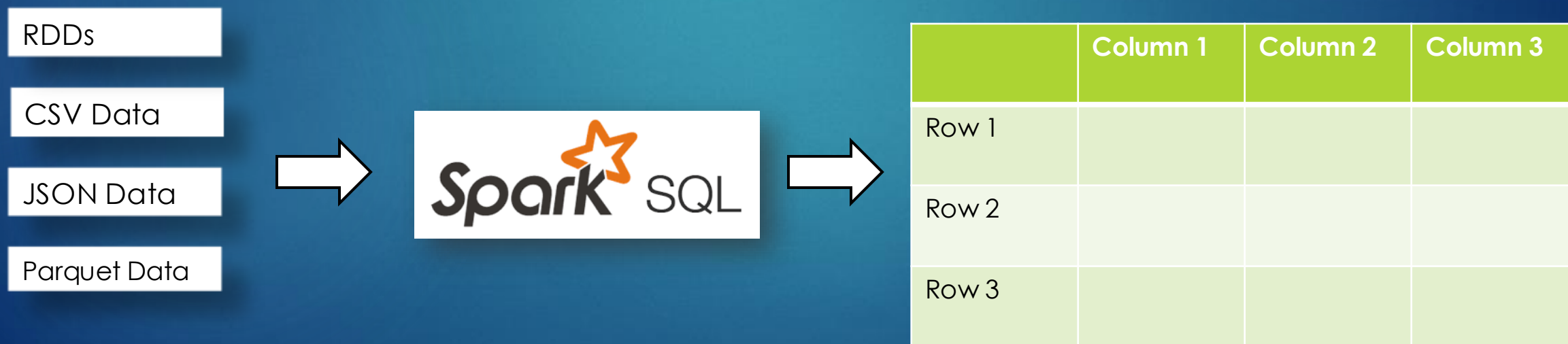
- ▶ Spark SQL uses a cost-based optimizer, code generation, and columnar storage to execute queries efficiently across thousands of nodes
- ▶ It provides structured interfaces that give Spark more insight into both the data and the computations being performed
- ▶ Spark SQL uses this additional information to apply advanced optimizations that improve query performance
- ▶ As a result, Spark SQL can run queries up to 100 times faster than Hadoop MapReduce



Spark SQL - DataFrames

28

- ▶ A DataFrame in Spark is a distributed collection of data organized into named columns, similar to a table in a relational database
- ▶ DataFrames support operations such as filtering, grouping, and aggregations, and they integrate seamlessly with Spark SQL for querying
- ▶ They can be created from structured data files (like CSV or JSON), existing RDDs, or external data sources such as relational databases



Spark SQL (DataFrame) – Simple Example

29

Input – /example/people.json

```
{ "name": "Michael" }  
{ "name": "Andy", "age": 30 }  
{ "name": "Justin", "age": 19 }
```

Create a DataFrame based on the content of a JSON file:

```
df = spark.read.json("examples/people.json")
```

Read JSON file into a DataFrame

```
df.show()
```

Displays the content of the DataFrame

```
# +----+-----+  
# | age| name|  
# +----+-----+  
# |null|Michael|  
# | 30| Andy|  
# | 19| Justin|  
# +----+-----+
```

```
df.printSchema()
```

Print the schema in a tree format

```
# root  
# |-- age: long (nullable = true)  
# |-- name: string (nullable = true)
```

Spark SQL (DataFrame) – Simple Example

30

```
df.select("name").show()
```

Select only the "name" column

```
# +-----+
# |  name  |
# +-----+
# |Michael|
# |  Andy  |
# | Justin |
# +-----+
```

```
df.select(df['name'], df['age'] + 1).show()
```

Select everybody, but increment the "age" by 1

```
# +-----+-----+
# | name| (age + 1)|
# +-----+-----+
# |Michael| null|
# |  Andy  |  31|
# | Justin |  20|
# +-----+-----+
```

```
df.filter(df['age'] > 21).show()
```

Select people older than 21

```
# +---+-----+
# |age| name|
# +---+-----+
# | 30| Andy|
# +---+-----+ #
```

Spark SQL (Native SQL) – Simple Example

31

- ▶ Running native SQL queries is also possible

```
sqlDF.createOrReplaceTempView("people")
```

← Register the DataFrame as a SQL temporary view

```
sqlDF = spark.sql("SELECT * FROM people")
```

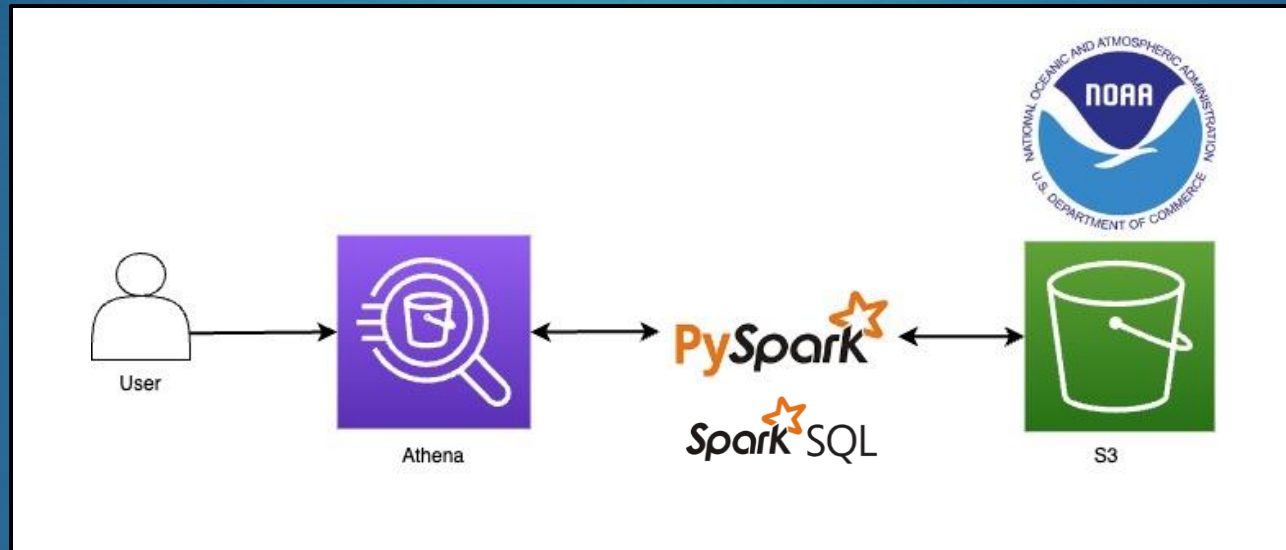
← Use the typical SQL Query syntax (SELECT...WHERE, etc.)

```
sqlDF.show()
```

```
# +-----+-----+
# | age| name|
# +-----+-----+
# |null|Michael|
# | 30|  Andy|
# | 19|  Justin|
# +-----+-----+
```

Data Exploration & Visualization

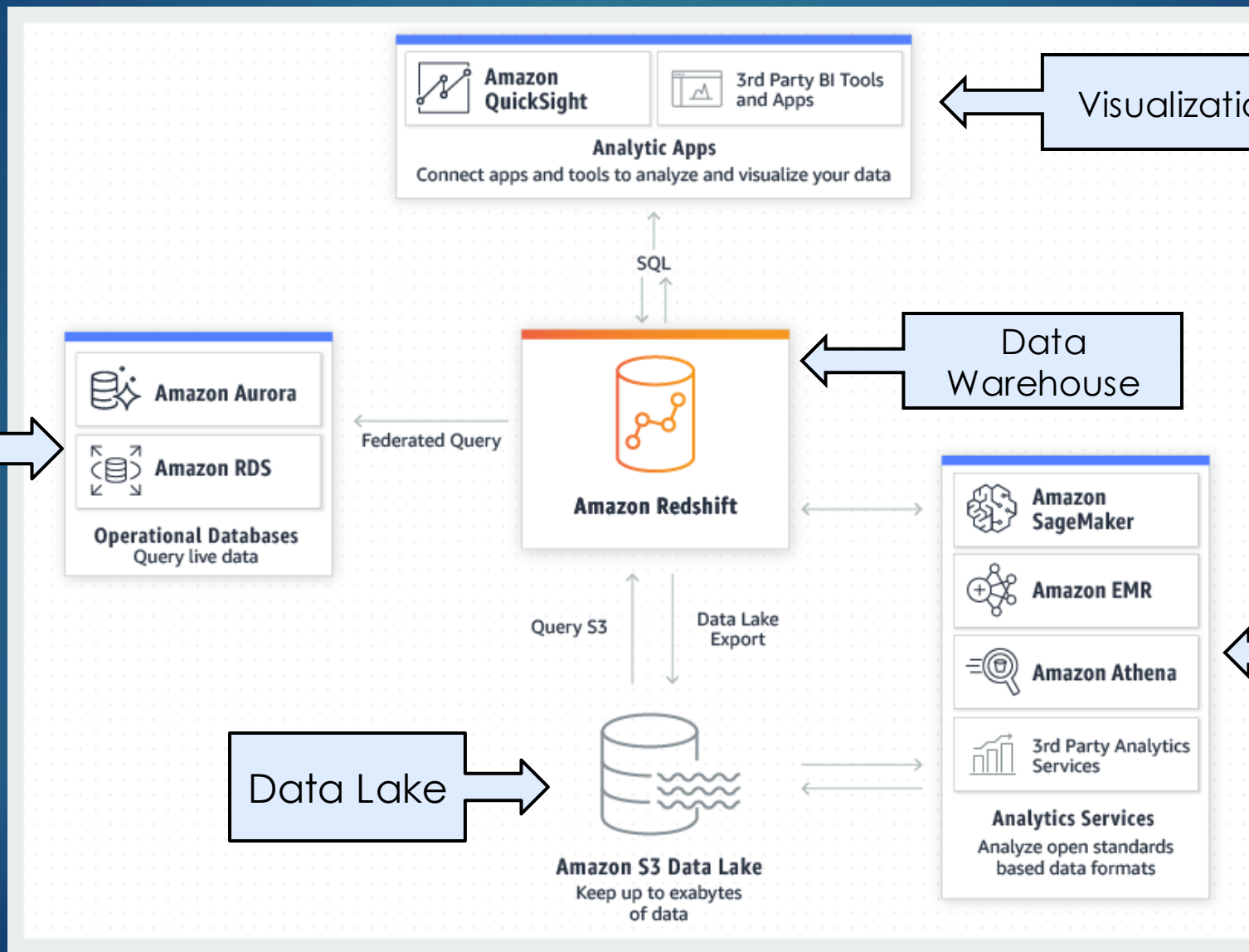
- ▶ You will be using Athena a serverless, interactive analytics service to query [NOAA Global Surface Summary of the Day](#) using both Spark SQL and SQL
- ▶ This data contains weather related information from over 9000 weather stations



- ▶ Activity is located in [Assignments > Activity - Data Exploration & Visualization](#)
- ▶ Opportunity to earn another bonus point

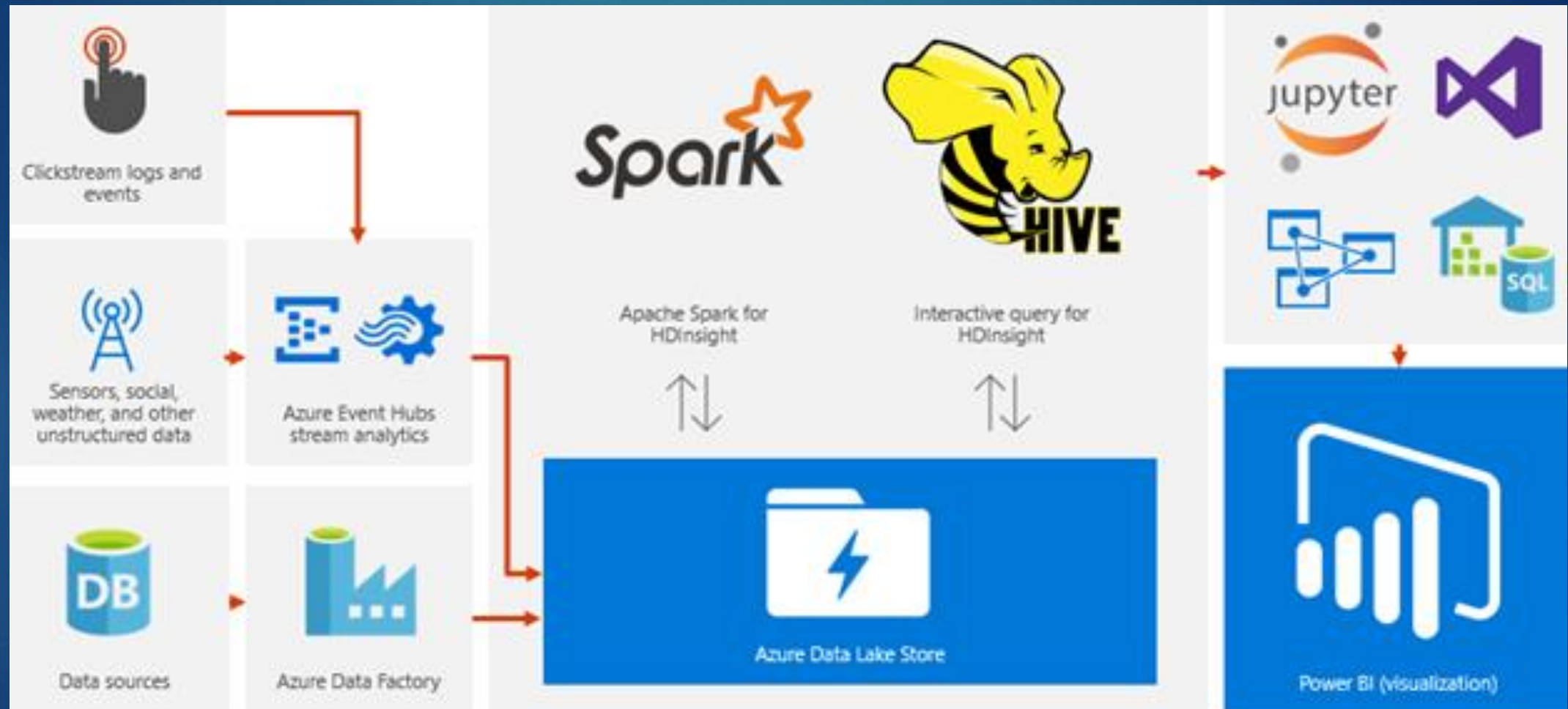
Tying it all together on AWS

33



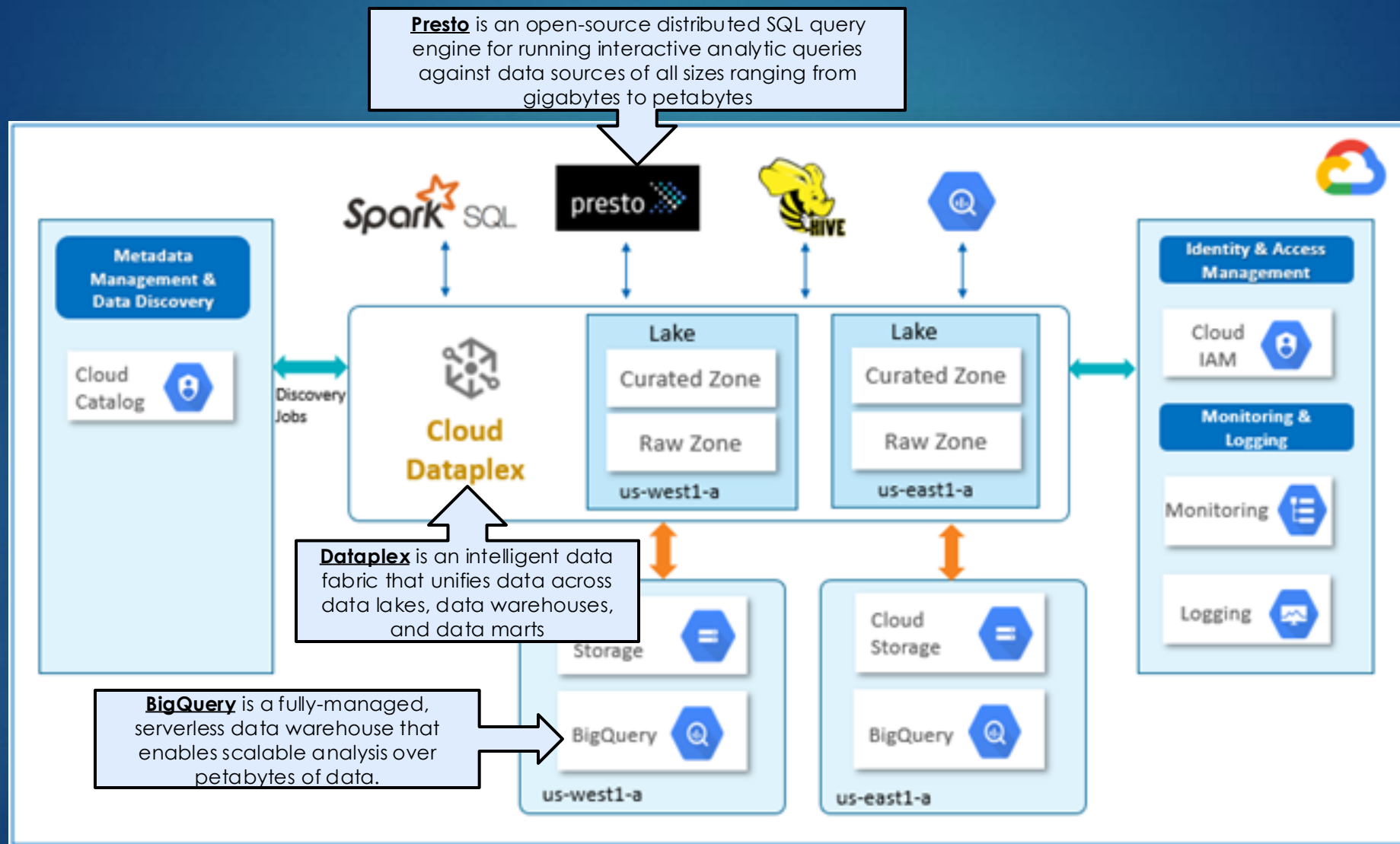
A Data Lake solution on Azure

34



A Data Lake solution on Google Cloud

35



Big Data and Data Science

36

- ▶ Data Science and Cloud Computing work closely together, as the cloud provides scalable infrastructure for storing, processing, and analyzing massive datasets
- ▶ Data Scientists analyze diverse types of data stored in the cloud, including structured, semi-structured, and unstructured data from sources such as IoT devices, social media, emails, and enterprise systems
- ▶ Their work involves cleaning, processing, modeling, and interpreting data to generate insights and actionable strategies for organizations
- ▶ With the exponential growth of Big Data, organizations increasingly rely on cloud platforms and skilled Data Scientists to unlock value through advanced analytics, machine learning, and AI

The world is generating data at a higher rate, and so the need of "Data Science" & "Data Analytics" tools increases to analyze and manage this "Big Data".

WHAT IS DATA SCIENCE?	WHAT IS DATA ANALYTICS?	WHAT IS BIG DATA?
Data Science is a field that refers to the collective processes, theories, concepts, tools and technologies that enable the review, analysis and extraction of valuable knowledge and information from raw data.	Data Analytics (DA) is the process of examining data sets in order to draw conclusions about the information they contain, increasingly with the aid of specialized systems & software.	Big Data refers to voluminous amounts of structured or unstructured data that organizations can potentially mine & analyze for business gains.
APPLICATION AREAS		
1. Digital advertisements 2. Internet Research 3. Recommender System 4. Image/Speech Recognition	1. Gaming 2. Travel 3. Energy Management 4. Healthcare	1. Communication 2. Retail 3. Financial services 4. Education
TOOLS & LANGUAGES		
1. Python 2. SAS 3. SQL	1. R 2. Tableau Public 3. Apache Spark	1. Hadoop 2. NoSQL 3. Hive
ANNUAL SALARY		
Data Scientist \$130,323	Big Data Specialist \$69,845	Data Analyst \$62,066



- ▶ Going back to your scenarios one last time, your team is tasked to analyze opportunities for Big Data and Analytics
- ▶ Questions to answer:
 - ▶ What type of analytics can be created and what data would be needed? Is a Data Lake required? What about a Data Warehouse?
 - ▶ Based on the data you are working with; what type of Big Data technologies would be recommended to process the data?
 - ▶ How often would you process this data?
 - ▶ What is the value of the analytics you would be providing (e.g. increased sales)
- ▶ Submit completed template to **Assignments > Activity - Big Data Analytics Recommendation**
 - ▶ Only 1 submission per team
 - ▶ The scenarios can be found in **Assignment > Activity - Cost Estimating Scenarios**